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Enhancing Students' Environmental Awareness of Climate Change by Incorporating the Metaverse Into Educational Settings

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Abstract: The rapid advancement of educational technologies has created new opportunities to integrate language learning with education of global sustainability. This study explored how the metaverse could help enhance students' environmental awareness of climate change via the perspectives of instructors teaching English as a foreign language (EFL). Specifically, it examined instructors' knowledge of the metaverse, their perceptions of its educational potential, and the challenges associated with its implementation. A mixed-method research design was employed using an online questionnaire completed by 70 EFL instructors from Kosovo, Montenegro, North Macedonia, Albania, Turkey, Poland, and Saudi Arabia. Quantitative data were analyzed using descriptive statistics and a Chi-square test of independence, while qualitative responses were examined through inductive thematic analysis. The findings indicated that most participants considered environmental education a key component of English language teaching and viewed the metaverse as a promising tool for promoting awareness of climate change. Nevertheless, some instructors reported limited familiarity with the technology and identified barriers like insufficient digital infrastructure, infringement of data privacy, unequal access to technological resources, and the need for professional development. The Chi-square analysis further revealed a statistically significant relationship between teaching experience and perceptions of the metaverse, with less experienced instructors expressing greater optimism regarding its educational potential than their more experienced counterparts. The findings suggested that although EFL instructors demonstrated strong willingness to integrate immersive technologies into language education, successful implementation would depend on targeted teacher training, improved technological infrastructure, and institutional support.

Keywords: Metaverse; Environmental education; English as a foreign language; Virtual worlds; Climate change

1. Introduction

In the 21st century, the rapid evolution of technology has brought unprecedented challenges and opportunities to education. The metaverse has emerged among these advances as a new and inclusive digital space with considerable potential to change and transform traditional educational practices. This study explored the perspectives of instructors teaching English as a foreign language (EFL) on how to integrate the metaverse into language teaching to enhance students' environmental awareness and their understanding of climate change. Climate change has become a serious and problematic global phenomenon, a critical issue that requires immediate responses and feasible solutions. The impact of climate change is increasing; it affects all spheres of life in the environment, society, and economy, and as a result, this phenomenon underlines the necessity of effective education that fosters a better understanding of climate-related issues among the younger generation.

Given the transformative potential of advanced technology in education, this study investigated how the metaverse could be used as a technological tool to improve environmental education within the context of English language teaching and learning. The metaverse, which is broadly defined as a shared virtual collective space, provides users with immersive and interactive experiences that exceed the limitations of traditional learning environments. As educational institutions consider incorporating digital technologies, using the metaverse as a platform for environmental education stands for a unique opportunity to engage students in the complexities of climate change.

The metaverse has received a lot of attention as both an advanced technology tool and a future educational trend. Since 2021, numerous scientific studies have been conducted on a global scale to spotlight its potential (Choi & Kim, 2017; Dwivedi et al., 2022; Gartner, 2022; Guo & Gao, 2022; Hwang & Chien, 2022; Park & Kim, 2022; Tlili et al., 2022; Zhang et al., 2022a). Offering an immersive experience of virtual reality (VR) and augmented reality (AR), the concept of the metaverse encompasses the merging of technology and reality. Individuals could engage in a variety of activities within this digital domain, such as education, work, art appreciation, commerce, gaming, musical performances, etc. Users can interact with others, foster connections, and create social connections through personalized digital representations, facilitating seamless communication, collaboration, and sharing of experiences (Kim et al., 2023). There are three main benefits of using metaverse applications when learning a foreign language. First, students can create avatars to represent themselves anonymously, allowing shy or introverted people to communicate more effectively. Second, the metaverse enables role-playing scenarios, allowing students to hone their speaking skills in dynamic and adaptable situations that mirror real-life discourse. Third, students can study remotely, collaborate with peers from other countries, and improve their communication skills for using other internet-connected services (Imaoka et al., 2012).

This study, by leveraging the advanced technology derived from the metaverse, spotlighted its potential to improve environmental education and promote effective communication in English language contexts.

1.1 Research Goal

The primary goal of the present study is to investigate the attitudes of EFL instructors toward incorporating the metaverse into classrooms, in order to raise students' awareness of climate change.

1.2 Research Objectives

The following are some of the objectives of this study: to assess English language instructors' awareness of the metaverse and its potential to raise awareness of climate change in language teaching; to examine changes in English language instructors' attitudes and beliefs regarding the integration of technology, specifically the metaverse, in environmental education for language teaching; and to identify and analyze the challenges that EFL instructors face.

1.3 Research Questions

RQ1: What is EFL instructors' current level of awareness of the metaverse, and how do they see its potential for raising awareness of climate change in language learning?

RQ2: What are the benefits of incorporating the metaverse into language instruction for climate change education, according to EFL instructors?

RQ3: What challenges do EFL instructors face when incorporating the metaverse into language instruction for climate change education, and what perceived benefits are associated with this approach?

1.4 Significance of the Research

The study stemmed from its potential to inform educational practices that link language teaching and environmental awareness. Understanding how English language professors perceive the integration of the metaverse in their classroom settings could inspire educational policymakers, curriculum developers, and educators with valuable insights into the feasibility and effectiveness of using digital environments to address global challenges. Furthermore, this study added to the larger discussion about the interplay of language education, technology, and environmental awareness, with implications for pedagogical innovation in the pursuit of a more sustainable and informed future.

1.5 Research Gaps

While a growing body of literature assessed the general possibilities of the metaverse in foreign language acquisition, such as improving speaking skills through avatars or facilitating remote collaboration, there is a critical lack of empirical research examining its specific application in global sustainability education. Current studies emphasized technical frameworks or general vocabulary gains, leaving a notable gap regarding how inclusive spaces could be leveraged to resolve complex socio-scientific issues such as climate change within language curricula. Furthermore, existing research rarely captured the collective knowledge of EFL educators, spanning multiple geographically and culturally diverse countries. This study filled these gaps by providing a cross-national empirical assessment of instructors' knowledge, perceived pedagogical opportunities, and systemic challenges related to integrating the Metaverse for environmental awareness

2. Literature Review

The metaverse is a virtual reality space that could be accessed by an infinite number of users simultaneously and indefinitely. It features a scalable and interoperable network of real-time rendered 3D virtual worlds that ensure data continuity, including identity, history, entitlements, objects, communications, and payments. The metaverse provides a platform for users to obtain a sense of presence and identity (Kern, 2024). Experts in the metaverse defined it as an ecosystem that included AR, artificial intelligence (AI), and other innovative technology (Kern, 2024). When combined, they will form the next generation of the internet, resulting in a more immersive, or spatial online experience and a seamless merging of the real and virtual worlds. A standard development organization called XRSI has collaborated with subject matter experts to establish a comprehensive definition which enumerates the cutting-edge technologies they believe will be significant to the metaverse: a system of linked virtual worlds with the following essential qualities encompassing interoperability, presence, persistence, and immersion. Several convergent technologies including enhanced computer graphics, hardware, and network capabilities, extended reality (XR), AI, decentralized ledger technologies (DLTs), neurotechnology, optics, and biosensing technologies, to enable the metaverse, in other words the next generation of the Internet.

According to the Google Trends index, the term "metaverse" grew in popularity in 2022. Stephenson (1992)'s book *Snow Crash* in 1992 first mentioned the term. This science fiction laid the groundwork for the concept of the metaverse, inspiring early developers to create platforms that were compatible with it. It also influenced the development of systems such as Adobe Connect, Microsoft Teams, and Zoom, which aim to improve virtual engagement, communication, and collaboration (Kim et al., 2023). The metaverse encourages interaction, communication, and cooperation, to offer an ideal environment for language acquisition (Aydın, 2022; Kim et al., 2023). This virtual realm provides opportunities for language learners to learn in an interactive and communicative setting, as it necessitates social interactions among individuals. According to Aydın (2023) and Kim et al. (2023), learners could gain an understanding of both the language and its associated culture by being exposed to comprehensible input, engaging in meaning negotiation, and producing language output. Students could improve their critical thinking and problem-solving skills by learning and producing language. Actively participating in conversations encourages students to study independently and collaborate with others during their language learning journey. The metaverse can provide a learner-centered environment that supports both independent and collaborative language learning (Aydın, 2023; Kim et al., 2023). Through immersive experiences, it may also enhance learners' language awareness, problem-solving skills, and motivation. It demonstrated EFL teachers' critical role in developing interactive metaversal activities and providing feedback, as well as addressing security and privacy concerns. The study emphasized the significance of teachers' technical skills, positive attitudes, and perceptions in successfully incorporating the metaverse into language education.

2.1 Metaverse Virtualization

The term "metaverse", which refers to a novel virtual realm that extends beyond our physical world, is derived from a combination of "meta", which means transcendence, and "verse", which represents the entire cosmos or global sphere. The term "metaverse" first appeared in the cyberpunk science fiction masterpiece, *Snow Crash*, written by acclaimed American author Neal Stephenson in 1992 (Joshua, 2017; Stephenson, 1992). According to scholarly sources, the metaverse has exciting potential for the future, particularly in the field of education. It is proposed that combining the metaverse-related technologies with elements from both virtual and physical educational environments could improve the learning experience. According to researches (de la Fuente Prieto et al., 2022; Rospigliosi, 2022; Suzuki et al., 2020), the metaverse could serve as a novel educational environment.

2.2 Educational Framework of the Metaverse

Kang (2021) described the metaverse paradigm as comprising hardware, computing, networking, virtual platforms, interchange tools, standards, payment services, content, services, and assets. Park & Jeong (2022) grouped the metaverse into three major components—hardware, software, and content—and identified user interaction, implementation, and use as three broad dimensions of the metaverse. Hwang & Chien (2022) further examined its educational applications from an artificial intelligence (AI) perspective, highlighting the roles of intelligent peers, tutees, and tutors in supporting educational services. Because the metaverse integrates multiple technologies rather than functioning as a stand-alone technology, the present study adopts a holistic framework for its implementation in education, informed by these earlier models and the synthesis provided by Zhang et al. (2022b).

2.3 Potential Future Applications of the Metaverse in the Field of Education

Facebook's virtual announcement of its rebranding as Meta demonstrated several potential applications of the metaverse, including gaming, work, and learning (Meta, 2021). In education, the metaverse may reduce temporal and geographical constraints for both learners and educators. Its immersive and interactive features could also enable learners to perceive, explore, and create in ways that are difficult to reproduce in conventional settings. These affordances suggest that the metaverse may provide an innovative environment for future education. The following discussion therefore considers several of its potential educational applications, while recognizing that other possibilities may emerge as the technology develops.

2.4 Enhancement of Language Teaching

The metaverse is a useful tool for learning and improving language skills. Since the beginning of the twenty-first century, it has become increasingly important to be fluent in multiple languages in K–12 education, college, and the professional sphere. However, the traditional approach to language learning was comparatively passive, both inside and outside of the classroom, for a variety of reasons, including a lack of immersive practice and engagement opportunities (Liang et al., 2023). Scholars such as Guo & Gao (2022), Lee & Jeong (2022), Park & Jeong (2022), Parmaxi (2023), and Ryu (2022) believed the metaverse had significant potential for facilitating language acquisition. The metaverse has been used in language learning for numerous reasons. An immersive learning environment is initially critical for language acquisition, particularly during speaking and listening sessions (Chen et al., 2011; Lin & Wang, 2021). For instance, the goal of one communication skills lesson is to proficiently acquire the ability to engage in a dialogue about the process of obtaining flight information in an airport setting. It is extremely difficult for educators to arrange for an entire cohort of students to visit the airport, let alone invite airport personnel to visit the educational facility. In contrast, within the metaverse, students could engage in a variety of educational activities, such as simulated role playing and dialogue exercises, in the company of avatar companions or predetermined intelligent and non-playable characters who serve as air hostesses or airport personnel. Individuals looking to improve their language skills would find themselves in an authentic and meticulously created learning environment via the metaverse, thus allowing them to fully immerse themselves in the language acquisition process and bolster their linguistic capabilities alongside fellow students and instructors. Furthermore, it is critical to recognize that language proficiency requires consistent and ongoing practice outside traditional classroom settings. Non-native English speakers, for example, may face a challenge while studying EFL over the summer. Conversational exercises that involve speaking, listening, and translation could help them improve their English language skills. Learners may ask a fellow participant to join a virtual reality environment via an avatar, thus allowing them to practice conversations in a pre-arranged or recreated setting remotely. Given the capabilities of virtual assistants, learners could also benefit from the presence of intelligent language peers who provide language practice in the absence of their assigned partners. In this case, language learners may find virtual reality an appropriate platform for practising language skills more engagingly and interactively, both within and outside of the confines of a traditional classroom environment, which may also aid in the effective transfer of language ability to the real world (Zhang et al., 2022b).

2.5 Difficulties in Using the Metaverse in Education

As we explored the metaverse, we should be aware of the various risks that may arise because of this virtual space, as well as the potential impact on our educational endeavors and broader goals. In the following discussion, we would emphasize four key issues related to the intersection of the metaverse and education. To access the metaverse universe, educators and students should have a high-quality and reasonably priced intelligent wearable device (Parmaxi, 2023). Notably, important progress has been made in hardware, such as head-mounted displays (HMDs), though further improvements are still needed. It is worth noting that prolonged use of wearable devices may cause dizziness, blurred vision, cybersickness, and, in severe cases, physical falls (Tlili et al., 2022; Weech et al., 2019; Xi et al., 2023). Ultimately, this poses a potential safety risk in real-world scenarios. Furthermore, the cost of such hardware is an important consideration, as some people still regard these devices as prohibitively expensive (Taylor & Soneji, 2022). The security and protection of user data and privacy are critical, whether in a three-dimensional virtual environment or on the two-dimensional Internet. Data is important in governing the metaverse because it allows the collection of more specific information from users, such as transactions, consumption records, physical indicators (such as heart rate, blood pressure, health conditions, and so on), facial photographs, and other relevant data (Kye et al., 2021; Lv et al., 2022; Zhao et al., 2022). Furthermore, because of the increased level of online anonymity in the metaverse, people with little social experience, such as students, are more likely to encounter criminal activity, such as fraud, espionage, and unauthorized disclosure of information. Such unfortunate incidents constitute a direct violation of learners' privacy and may have negative consequences in their daily lives.

Educational materials and student work may also be copied or misused in the metaverse. Appropriate legislation, identity-verification mechanisms, and regulatory oversight will therefore be needed. Intellectual property and creative outputs could be tracked and verified using technologies such as blockchain, non-fungible tokens (NFTs), and cryptocurrencies (Berg et al., 2019; Thomason, 2022; Vidal-Tomás, 2022). Without adequate safeguards, the metaverse could become a poorly regulated space in which learners'rights and creative work are insufficiently protected. Its introduction into education also raises broader concerns related to security, ethics, and addiction. These issues require careful consideration because unresolved risks may outweigh the educational benefits. Future research should therefore examine how the capabilities of the metaverse can address existing educational limitations while minimizing its potential harms (Zhang et al., 2022b).

3. Methodology

3.1 Sampling Strategy and Participants

This study used a purposive non-probability sampling strategy to select participants who could provide informed knowledge in the intersecting areas of English language pedagogy and digital technology integration. The target population consisted of active EFL instructors. To ensure the relevance of the data, two clear inclusion criteria were set: (1) individuals must be actively teaching EFL at a secondary or higher education institution at the time of data collection; and (2) they must have at least one year of teaching experience. Potential participants who did not meet both conditions were excluded. The final sample included 70 validated EFL instructors spanning 7 countries: Kosovo, Montenegro, North Macedonia, Albania, Turkey, Poland, and Saudi Arabia. Participants were recruited through institutional professional networks, academic email lists, and regional associations of EFL instructors. A formal email invitation containing a secure link to the survey was distributed over a four-week period.

3.2 Instrument Development, Validity, and Reliability

The instrument for data collection was a self-administered and structured questionnaire, developed electronically via Google Forms. The structure of the questionnaire was divided into three distinct thematic blocks:

Block 1: Demographics: Included gender, age, country, and years of teaching experience.

Block 2: Perception and Knowledge: Assessed instructors' current knowledge of the metaverse and their views of its potential for environmental awareness using a 5-point Likert scale.

Block 3: Qualitative Insights: Contained open-ended questions targeting perceived benefits of implementation, challenges, and future suggestions.

To determine content validity, the initial draft of the instrument was reviewed by a panel of two senior faculty members specializing in educational technology and applied linguistics. Based on their expert feedback, the wording of the three Likert-scale questions was modified to enhance clarity and eliminate technical jargon. A pilot test was conducted with a subsample of five EFL instructors (excluded from the final 70) to confirm readability and basic stability. Since the instrument included exploratory and open-ended thematic segments, along with discrete perception items rather than a uniform psychometric scale, internal consistency was verified through interrater agreement during the qualitative coding phase. Standard Cronbach's alpha matrices were not adopted.

3.3 Procedures of Data Collection

Data collection was done entirely anonymously to mitigate social desirability bias. No personally identifiable information (such as names, IP addresses, or email accounts) was captured.

3.4 Analysis of Descriptive Statistic

This research adopted a descriptive survey design supplemented by qualitative content analysis. For the qualitative and open-ended sections of the questionnaire, textual data were processed using an inductive and systematic coding procedure. The raw text responses were first read line by line to identify recurring key words. These initial codes were then grouped into sub-themes based on conceptual similarities, which were finalized into main thematic frameworks by calculating clear frequency counts to establish priority and consensus across the sample.

4. Results

In this study, *n* denotes the number of responses for a given survey item or section. This section presents the empirical findings derived from the survey of EFL instructors across seven nations, with 70 participants in total (*n*

= 70). The data were structured to directly address the primary research questions and objectives of this study. The response frequency (*n*) varied across specific sections because the survey permitted multiple-choice selections for perceived benefits, while qualitative and open-ended feedback prompts remained optional.

Table 1. Demographic characteristics of the respondents

Variable	Category	Percentage (%)
Gender	Female	59.1
	Male	40.9
Age	Under 25	0.0
	25–34	22.7
	35–44	27.3
	45–54	36.4
	55 and above	13.6
Teaching experience	Less than 5 years	13.6
	5–10 years	27.3
	11–15 years	13.6
	16 years and above	45.5

Table 1 details the demographic profile of the participants based on gender, age, and teaching experience. The majority of the sample is female (59.1%) compared to male (40.9%). In terms of age, the largest group is 45–54 years old (36.4%), followed by 35–44 (27.3%) and 25–34 (22.7%), while those 55 and above make up 13.6%, and no participants are under 25. Additionally, the cohort is highly experienced, with nearly half having 16 years or more of teaching experience (45.5%), while 27.3% have 5–10 years, and the remaining participants are equally split between less than 5 years (13.6%) and 11–15 years (13.6%).

Figure 1 depicts the perspectives of EFL instructors on the importance of incorporating awareness of climate change into EFL education. The data showed that a significant majority of participants viewed the topic positively, with 40.9% rating it as “Very Important” and 13.6% as “Extremely Important”. In contrast, 27.3% of respondents considered it “Moderately Important”, while 18.2% considered it “Slightly Important”. Most notably, 0% of the instructors polled “Not at All Important”.

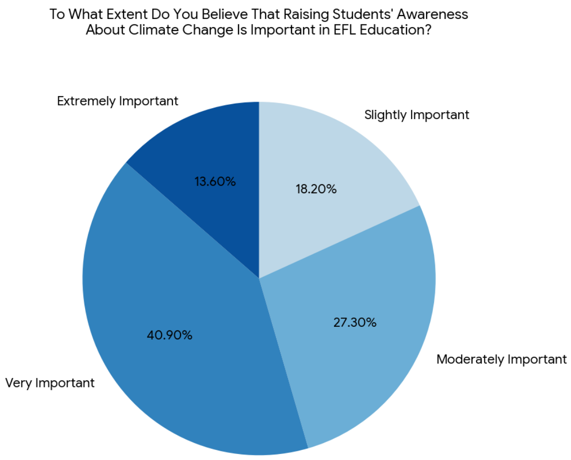

Figure 1. EFL instructors’ perception of the importance of climate change in English language education
Note: EFL = English as a foreign language.

Figure 2 illustrates EFL instructors’ degree of familiarity with the concept of the metaverse. According to the data in Figure 2, half of the respondents (50%) were only “Somewhat” familiar with the concept, while 40.9% reported being “Averagely Familiar”. At the extreme, 4.5% of the instructors were either “Not at All Familiar” or “Very Familiar” with the metaverse, with 0% identifying as “Extremely Familiar”.

Figure 3 depicts EFL professors’ perspectives on the extent to which incorporating the metaverse into classrooms could raise students’ awareness of climate change. Exactly half of the respondents (50%) believed it could do so “To a Great Extent”, while 36.4% said it could help “To a Moderate Extent”. The remaining 13.6% of the participants believed it would raise awareness “To a Small Extent”.

The data in Table 2 reveals that the vast majority of respondents (59 individuals) have a baseline experience level indicating they have never previously utilized the metaverse in an EFL classroom environment. Looking ahead, a smaller group of 10 respondents plans to integrate the metaverse for sustainability topics in future modules.

Meanwhile, only 1 respondent identified implementation challenges, specifically highlighting systemic barriers such as data privacy, student equity, digital literacy, and hardware accessibility.

As shown in Table 3, respondents identified several key benefits of integrating the metaverse into EFL classrooms for climate change awareness. The most frequent responses highlighted its potential for global collaboration and innovation (19), followed by viewing it as a step forward toward reversing climate change (12). It was also heavily noted for increasing moral and environmental responsibility (18 combined). Other benefits included raising interest and knowledge (8), driving behavioral change (5), and offering realistic, travel-free experiences (6 combined). Only a few emphasized saving the planet (1) or social-cultural awareness (1).

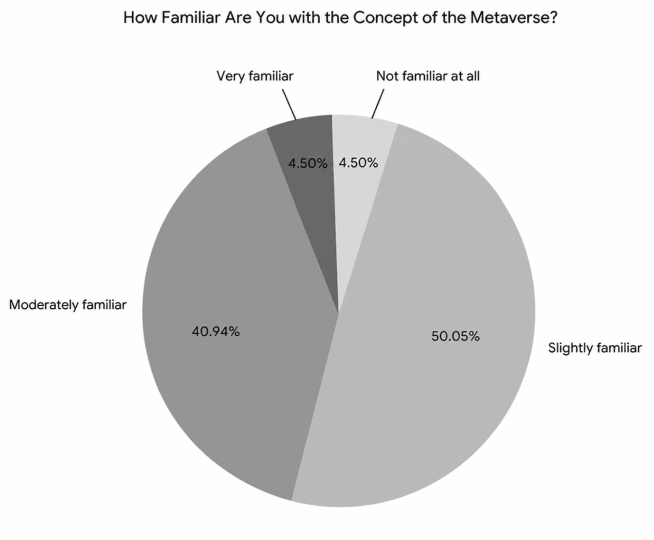

Figure 2. Survey of EFL instructors' familiarity with the metaverse

Note: EFL = English as a foreign language.

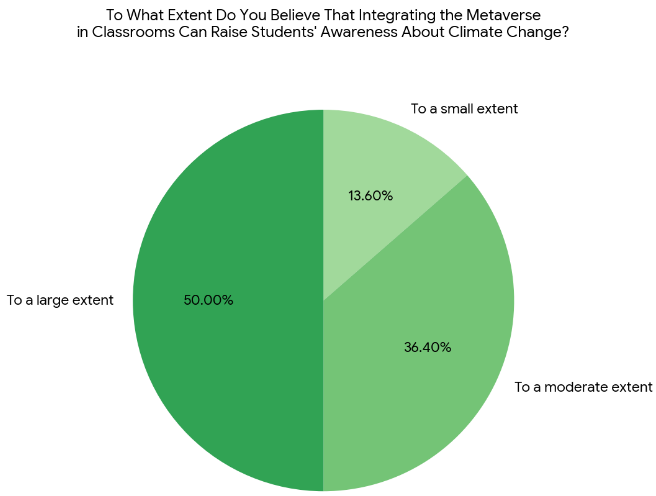

Figure 3. Implementation of the metaverse in classrooms for raising students' awareness of climate change

Table 2. Instructors' baseline experience and perceived challenges during implementation (n = 70)

Category	Sub-Theme	Respondent(s)
Baseline experience	Have never previously utilized the metaverse in an EFL classroom environment	59
	Plan to integrate the metaverse for sustainability topics in future modules	10
Implementation challenges	Systemic barriers (e.g., data privacy, student equity, digital literacy, and hardware accessibility)	1

Note: EFL = English as a foreign language.

Table 3. Potential benefits of integrating the metaverse into EFL classrooms for raising the awareness about climate change (n = 70)

Theme	Sub-Theme	Respondent(s)
Potential benefits of integrating the metaverse into EFL classrooms for raising awareness about climate change	A step forward, toward reversing climate change	12
	Global collaboration; innovation	19
	Save our planet	1
	Will add more moral responsibilities and obligations toward climate change	9
	Behavioral change	5
	Getting real-life experiences without the need to travel to different places	2
	Social cultural awareness	1
	It increases credibility because it is extremely realistic	4
	It could raise interest and increase knowledge about an important environmental problem	8
	Taking more responsibility to protect the environment and work for our well-being	9

Note: EFL = English as a foreign language.

Table 4 shows that 54 of the 70 participating EFL instructors responded to this open-ended question, while 16 did not. The most frequently mentioned recommendation among those who made suggestions was that the metaverse be used more frequently in EFL/English language teaching (ELT) instruction (32 responses). Instructors also emphasized the importance of incorporating the metaverse into coherent and effective teaching methods (8 responses) as well as learning more about how to use it (6 responses). Furthermore, four respondents emphasized the importance of teacher training in facilitating the successful integration of the metaverse. Less frequently, respondents proposed regular integration through monthly environmental classes in EFL (2 responses) and emphasized the role of the government and professors in providing access to the best available technology (1 response). One respondent stated that they had no suggestions. The findings indicate that EFL instructors prioritize more frequent integration of the metaverse, which is supported by appropriate pedagogical approaches, increased teacher knowledge, and professional development.

Table 4. Suggestions of metaverse integration in the classrooms

Theme	Sub-Theme	Respondent(s)
EFL instructors' suggestions for improving the integration of the metaverse in the EFL/ELT classrooms	The metaverse should often be incorporated into EFL/ELT	32
	The integration of the metaverse should be done regularly, with monthly environmental classes in EFL	2
	Teacher-Training is required for the integration of the metaverse	4
	The metaverse should be incorporated through coherent and effective teaching methods	8
	Government and professors/teachers should focus on bringing the best technology in EFL/ELT classrooms	1
	No suggestions	1
	To gain more knowledge on how to use the metaverse	6

Note: EFL = English as a foreign language; ELT = English language teaching.

4.1 Cross-Tabulation and Inferential Analysis

To evaluate whether instructors' teaching experience influenced their outlook on technological integration, a Chi-square (χ^2) test of independence was performed. Instructors were categorized by instructional tenure: Less Experienced (≤10 years) and Highly Experienced (≥11 years). Their perceptions regarding the metaverse's capacity to raise climate awareness were grouped into High Extent ("To a large extent") and Moderate/Low Extent ("To a moderate/small extent").

The Pearson Chi-square test showed in the Table 5 revealed a statistically significant relationship between teaching experience and perceived technological potential, χ^2 (1) = 3.94, p = .047 (p < 0.05). The data indicated that less experienced EFL instructors held significantly more optimistic views regarding the large-scale efficacy of the metaverse for environmental advocacy compared to their more veteran colleagues. This divergence highlighted the critical need for tailored professional development frameworks that addressed technological skepticism among senior faculty.

Table 5. Cross-tabulation: Teaching experience vs. perceived potential of the metaverse (n = 70)

Teaching Experience	High Extent	Moderate/Low Extent	Total
Less experienced (≤10 years)	18	10	28
Highly experienced (≥11 years)	17	25	42
Total	35	35	70

5. Discussion

The primary objective of this exploratory study was to evaluate the pedagogical utility of metaverse technology in fostering climate change awareness from the perspective of EFL educators. The empirical evidence demonstrates that immersive, inclusive digital environments are perceived by instructors to significantly enhance environmental education within language-learning paradigms. A critical finding of this research is the pronounced divergence between junior and senior faculty members regarding their approach to digital modalities. These distinct behavioral patterns align with established educational technology literature, which consistently identifies generational and tenure-based disparities in institutional technology adoption. For instance, the pervasive optimism observed among early-career instructors concerning spatial computing mirrors the empirical consensus established by previous researchers, who documented lower levels of systemic resistance to immersive tools among less experienced educators. These outcomes reinforce the necessity of implementing professional development and institutional support structures tailored specifically to faculty career stages. Conversely, the systemic challenges articulated by participants specifically concerning data privacy, digital equity, and hardware limitations confirm that infrastructure continues to serve as a primary impediment to adoption. This aligns with the critical framework articulated in recent literature, which posits that the pedagogical affordances of metaverse environments cannot be fully realized in language education absent standardized institutional support and equitable device distribution. By synthesizing environmental discourse with second language acquisition, this study extends the existing literature. It demonstrates that educators conceptualize immersive simulations not merely as instrumental tools for vocabulary acquisition but as affective spaces conducive to global civic engagement.

5.1 Strengths, Weaknesses, Opportunities, and Threats Strategic Analysis Framework

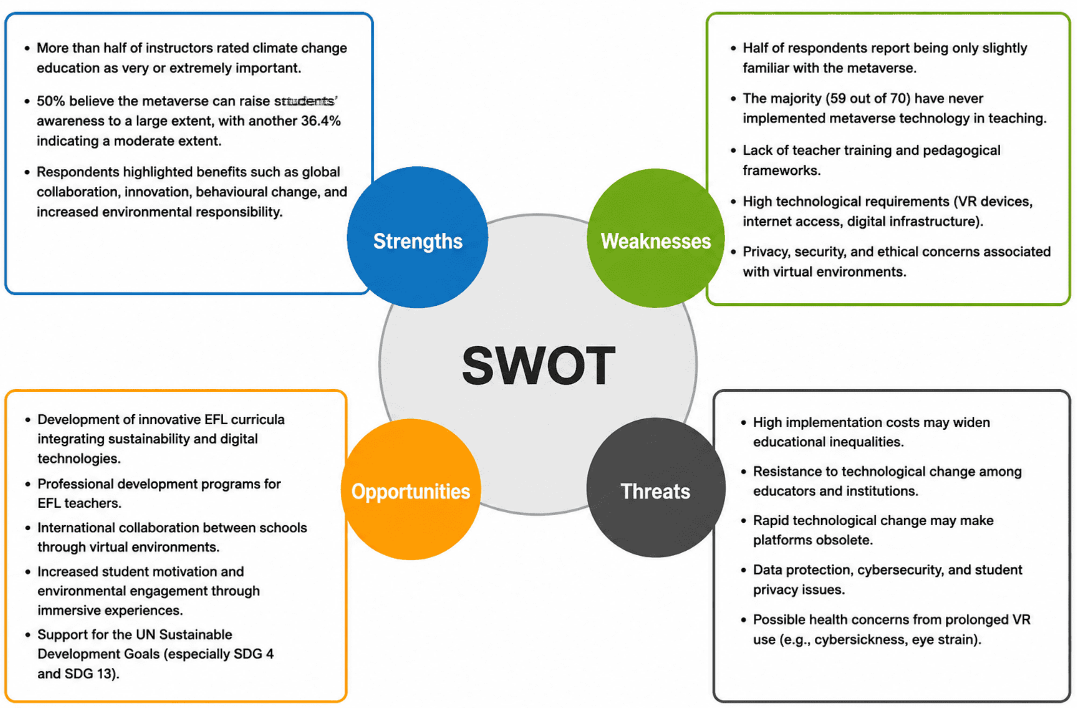

Figure 4. SWOT analysis of using the metaverse in EFL classrooms for climate change education
Note: EFL = English as a foreign language.

To translate the descriptive and qualitative findings into actionable institutional strategies, a Strengths, Weaknesses, Opportunities, Threats (SWOT) analysis framework was constructed as it is shown in Figure 4. It is essential to state clearly that this strategic matrix was not generated in isolation; rather, it was developed systematically through the direct integration of the empirical data with established frameworks found in the current educational technology and spatial informatics literature. By synthesizing the unique international insights of the 70 surveyed EFL instructors with existing peer-reviewed studies on inclusive technology infrastructure, this matrix provided a validated and complete assessment of the internal capabilities and external realities that governed the use of the metaverse for environmental language pedagogy.

6. Conclusions

This exploratory study investigated EFL instructors' perceptions regarding the integration of the metaverse technology to foster awareness of climate change. The empirical evidence suggested that educators viewed immersive and spatial platforms as highly capable environments for advancing sustainability education within language curricula. Rather than definitely demonstrating immediate classroom efficacy, the findings indicated a strong latent willingness among instructors to adopt these tools, provided that systemic barriers were dealt with. Specifically, the data pointed to a significant generation and experience gap in technological readiness, implying that future institutional deployments should arrange targeted professional development for senior faculty. Ultimately, while the implementation of the metaverse for environmental advocacy revealed promising pedagogical potential, its long-term success remains contingent upon resolving infrastructure disparities and establishing collaborative frameworks for cross-national teacher training.

Informed Consent Statement

Informed consent was obtained from all subjects involved in the study.

Data Availability

The data used to support the research findings are available from the corresponding author upon request

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Conflicts of Interest

The author declares no conflicts of interest.

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